What are ENM/SDM and why do we use them?

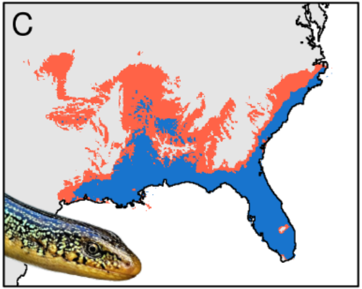
A brief history of SDMs

- 1807: Essay on the Geography of Plants (Humboldt)
- '10s-'20s: Grinnell & Elton
- '50s-'60s: Hutchinson, MacArthur, & Wilson
- '90s: Quantitative SDMs
- 2000s: SDMs as a recognized field
- 2010s: Boom of SDMs
- 2020s: Overflow of SDMs

The why of SDMs

- Fundamental theory
- Large-scale patterns
- Project future changes
- Conservation

The why of SDMs – Fundamental theory

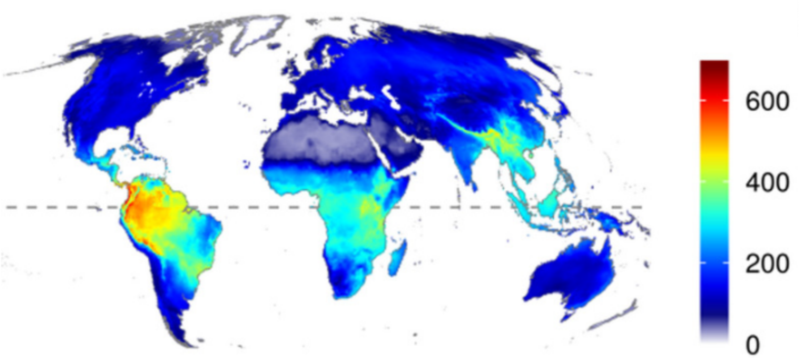


Suitable only with constant climate

Suitable even with variable climate

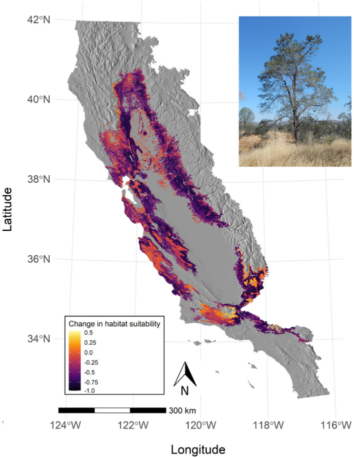
Effect of climate instability (Berti et al., 2025)

The why SDMs – Large scale patterns



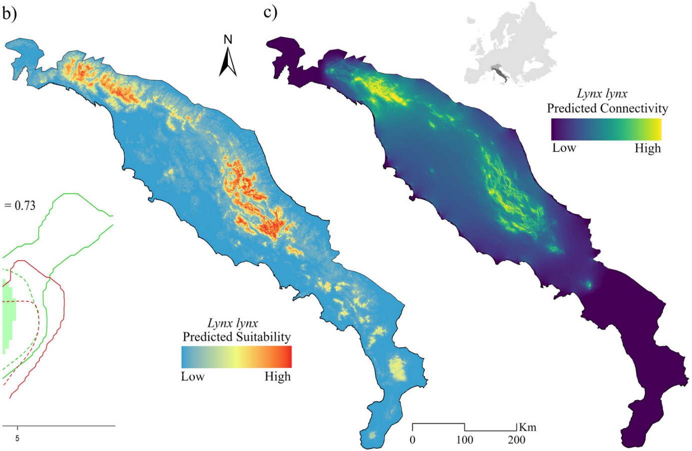
Vertebrate species richness (Biber et al., 2019)

The why SDMs – Project future changes



Change for *Pinus Sabiniana* (Franklin, 2023)

The why SDMs – Conservation

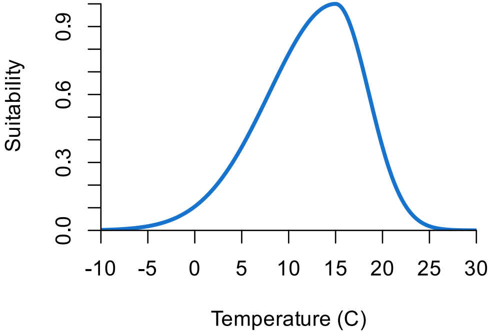


L. lynx (Serva et al., 2025)

The only theory you really need to know

Duality of SDMs

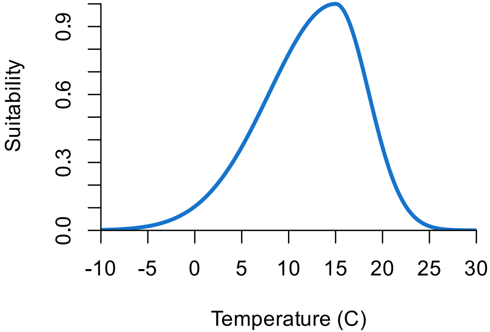
Ecological niche model (ENM)



Ecological space

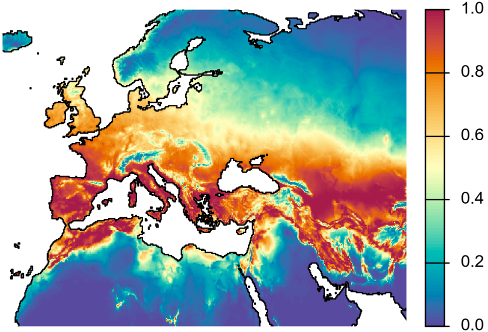
Duality of SDMs

Ecological niche model (ENM)



Ecological space

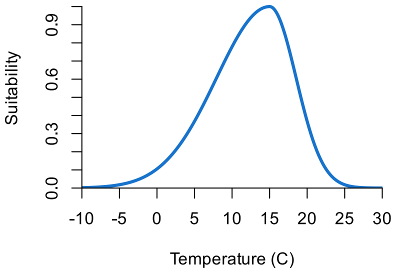
Species distribution model (SDM)

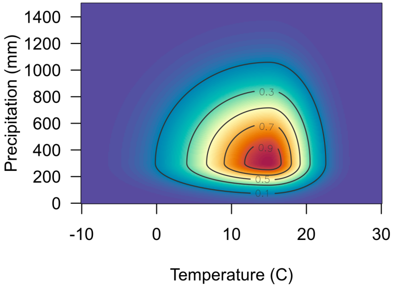


Geographic space

The niche of species

- **Convex**
  If a species tolerates two environmental conditions e_1 and e_2, it should tolerate a third $e_3 : e_1 \leq e_3 \leq e_2$
- **Multidimensional**
  Species generally respond to more than one environmental condition





One of many niches

Fundamental (Grinnellian) niche

$+$

Biotic (Eltonian) niche

$+$

Dispersal & meta-population niche

$=$

Realized niche

One of many niches

Fundamental (Grinnellian) niche

$+$

Biotic (Eltonian) niche

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Dispersal & meta-population niche

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Realized niche